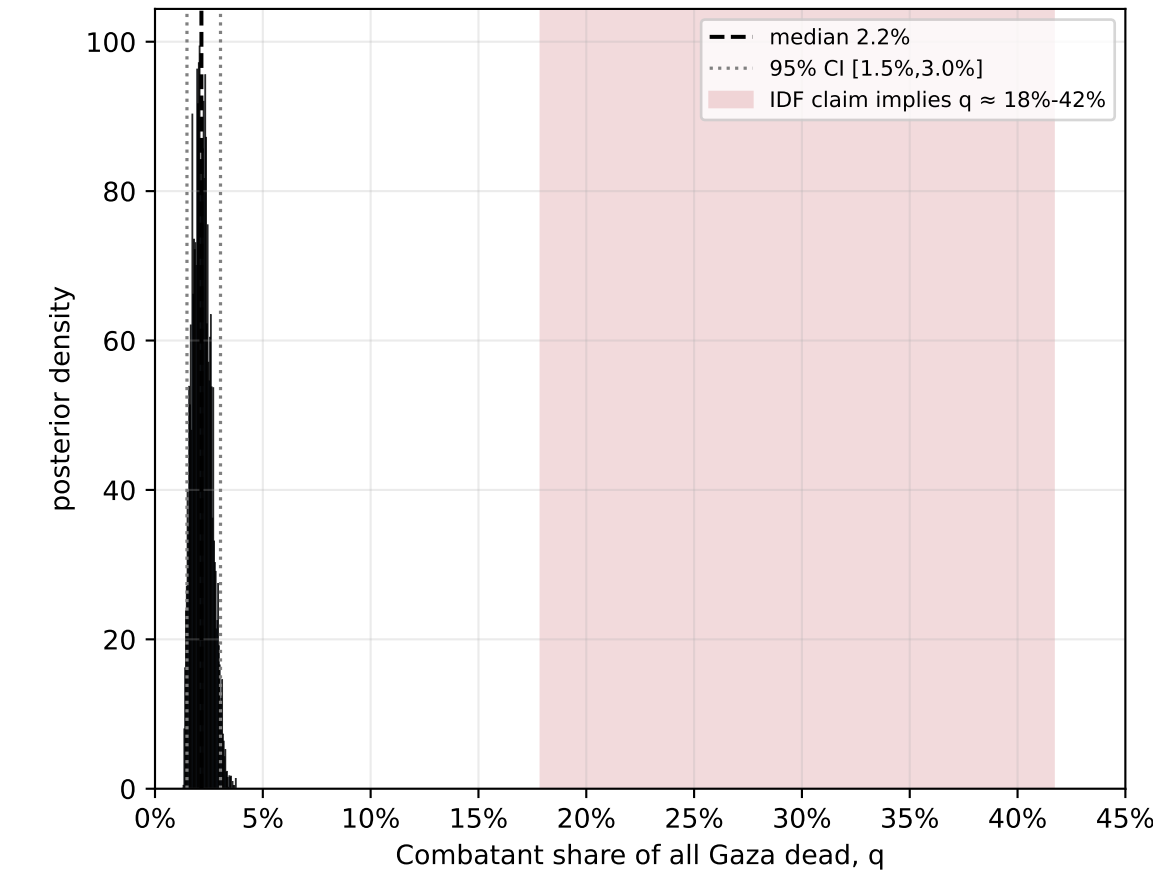


Gaza war — spatial Bayesian inference of militant vs civilian death share

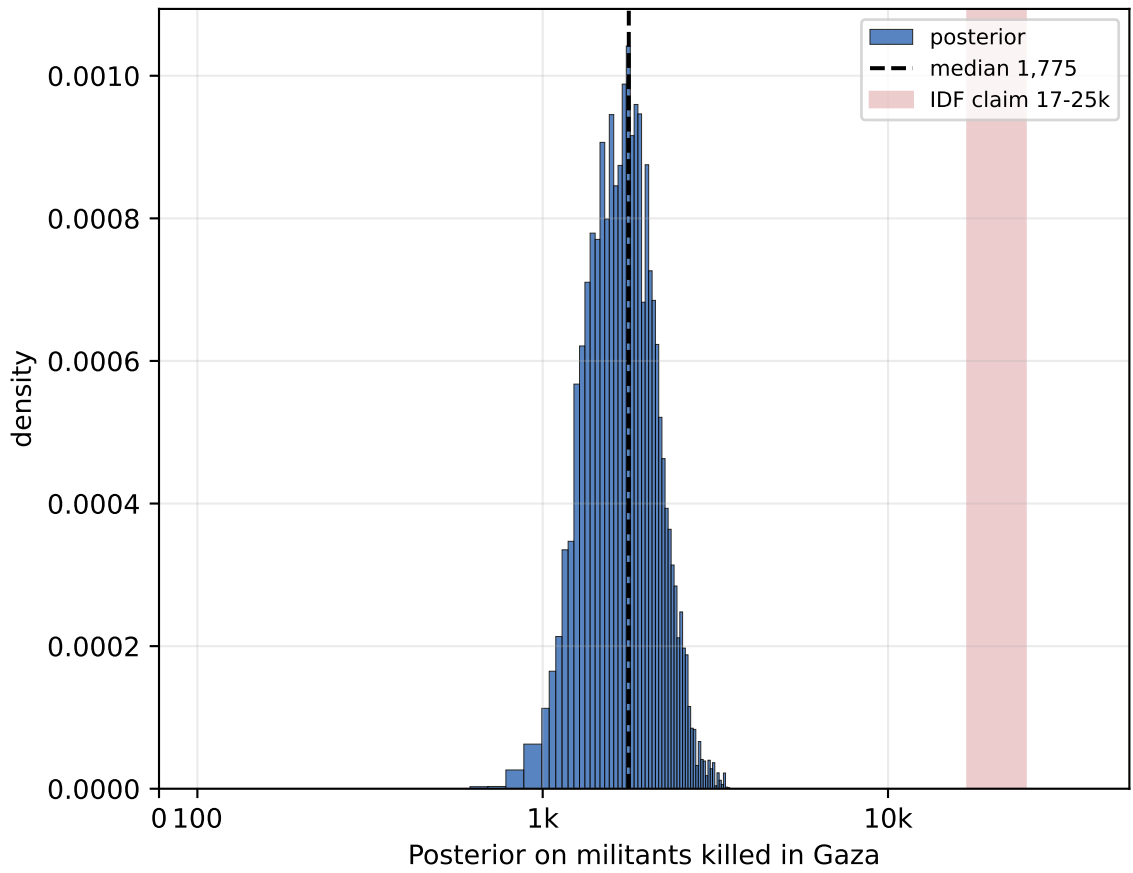
Forward model: 5 governorates × 80 cells; latent militant cell-density with clustering γ ; strikes target high-militant cells with bias α ; civilian adult-male exposure μ_M absorbs male over-representation.

Conditioned on PCBS demographics + MoH+OHCHR demographic split + MoH death tally + Lancet undercount. No belligerent claims used as priors.

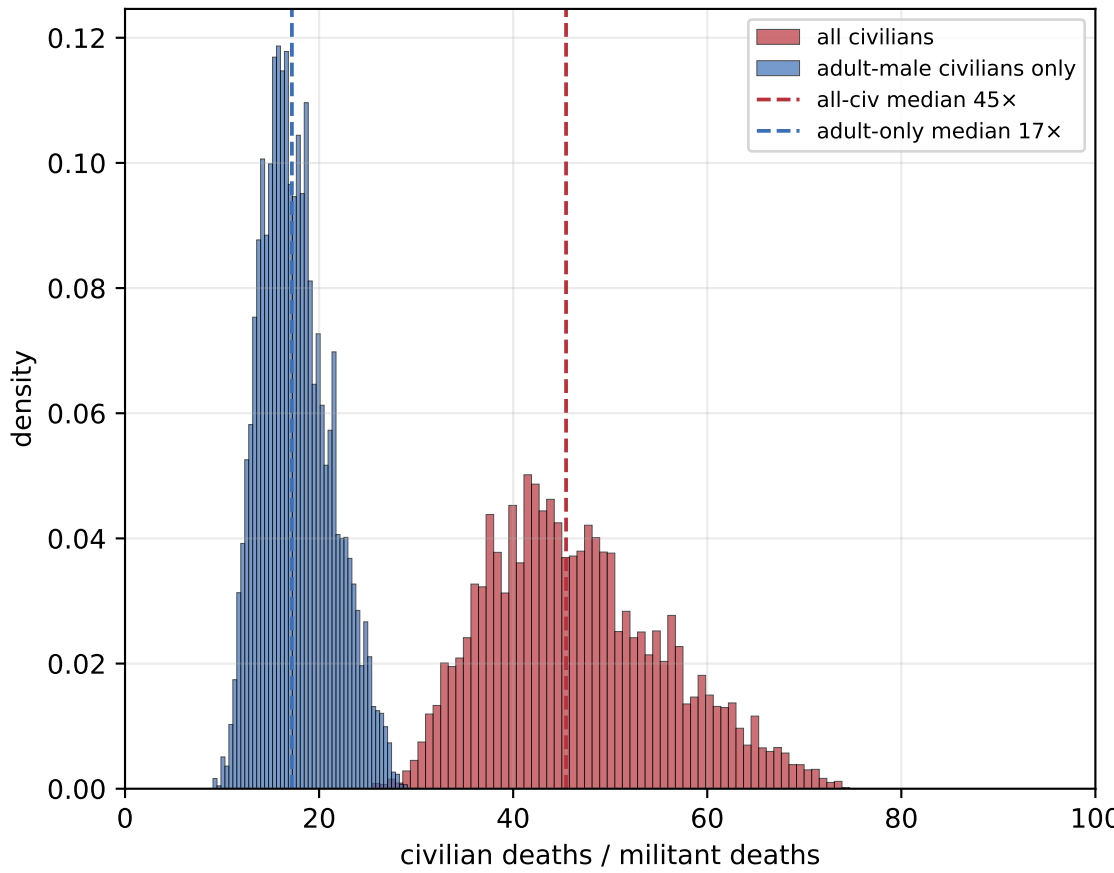
(1) Posterior on militant share q
median 2.2%, 95% CI [1.5%, 3.0%]



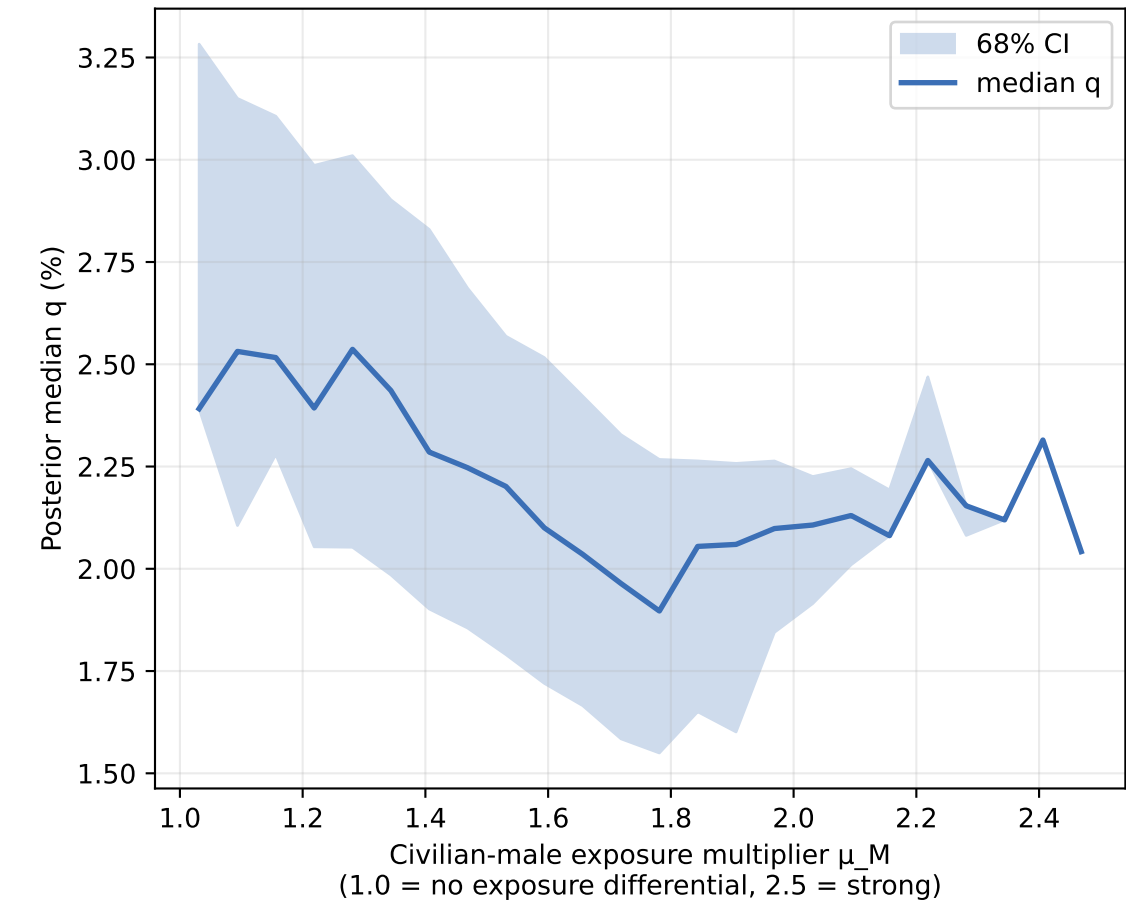
(2) Implied militants killed: median 1,775
95% CI [1,125, 2,684]



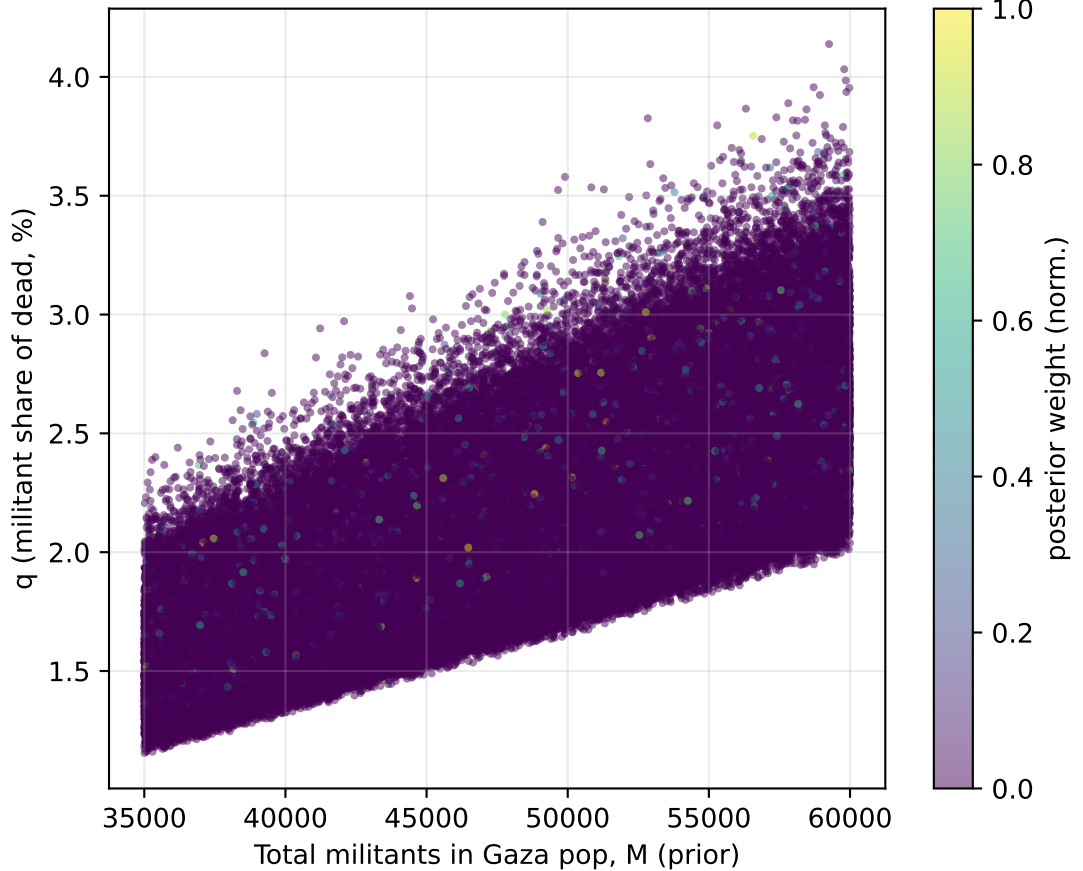
(3) Posterior on civ:mil ratio



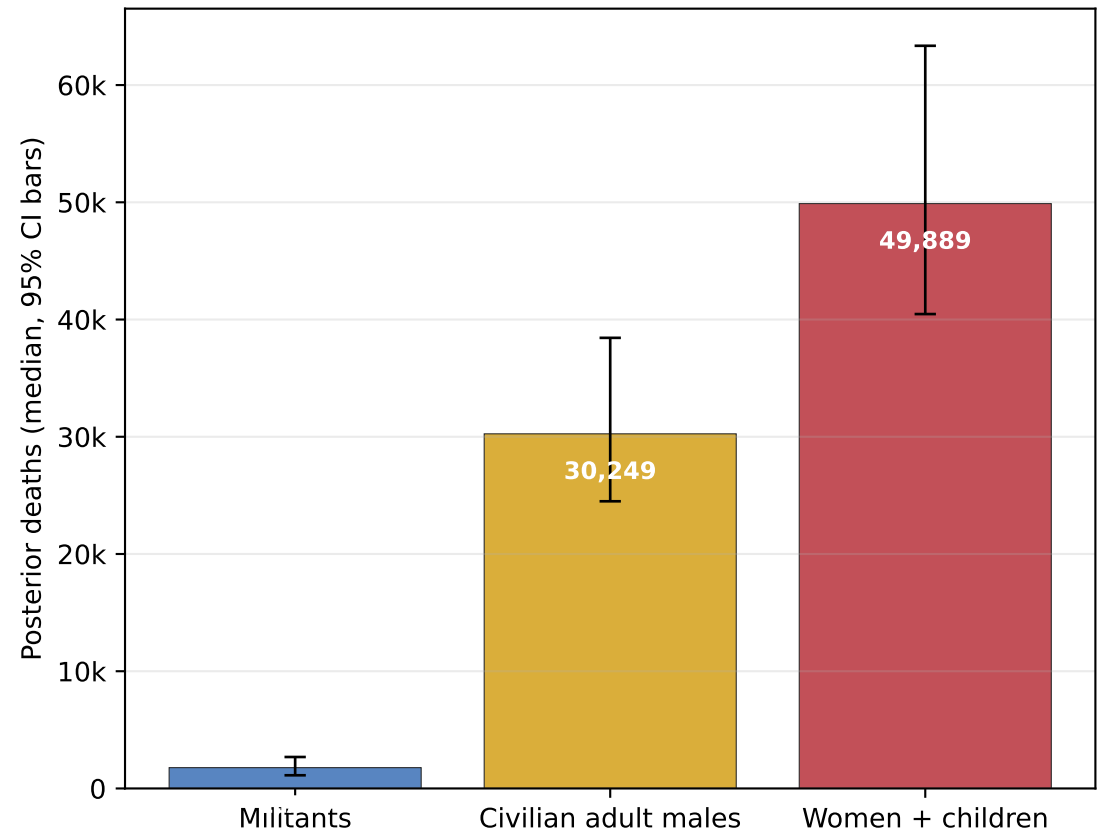
(4) Sensitivity: militant share q vs μ_M
(if civilian men were *not* over-exposed, q would rise)



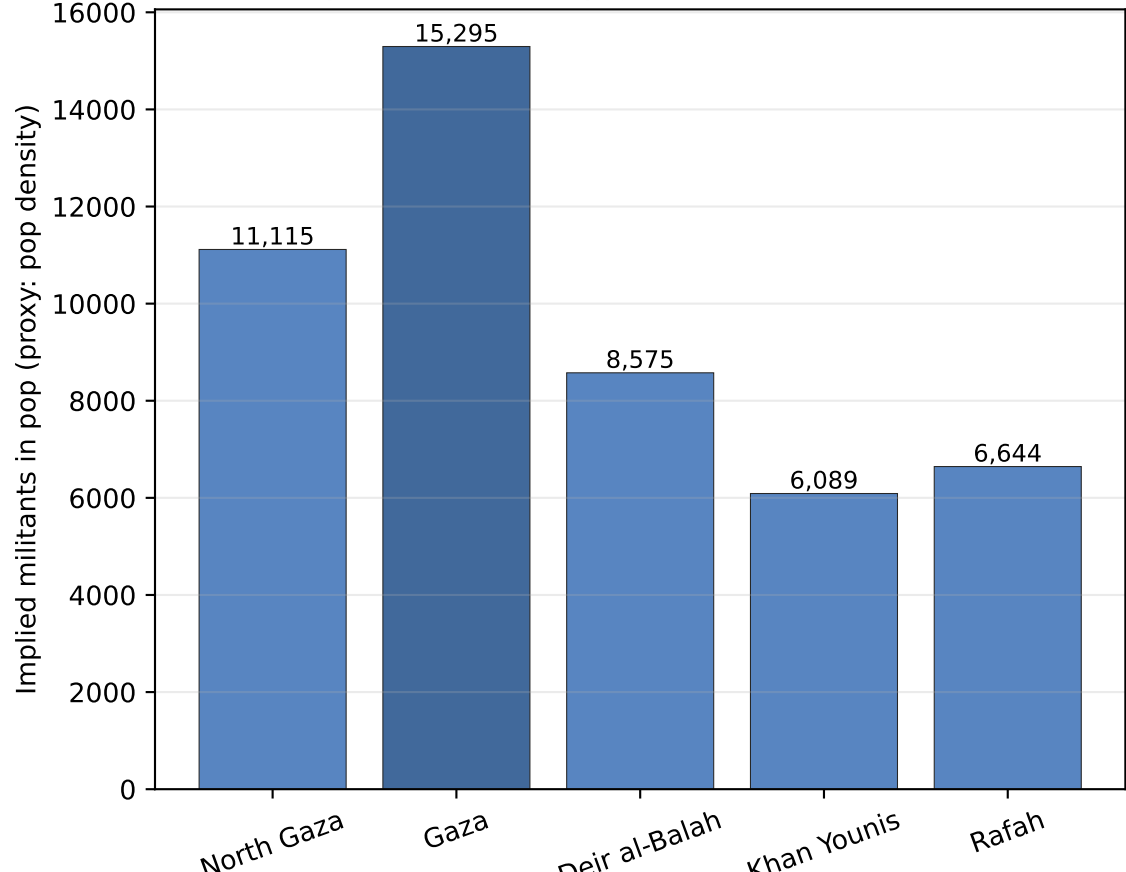
(5) Joint posterior over (M , q)



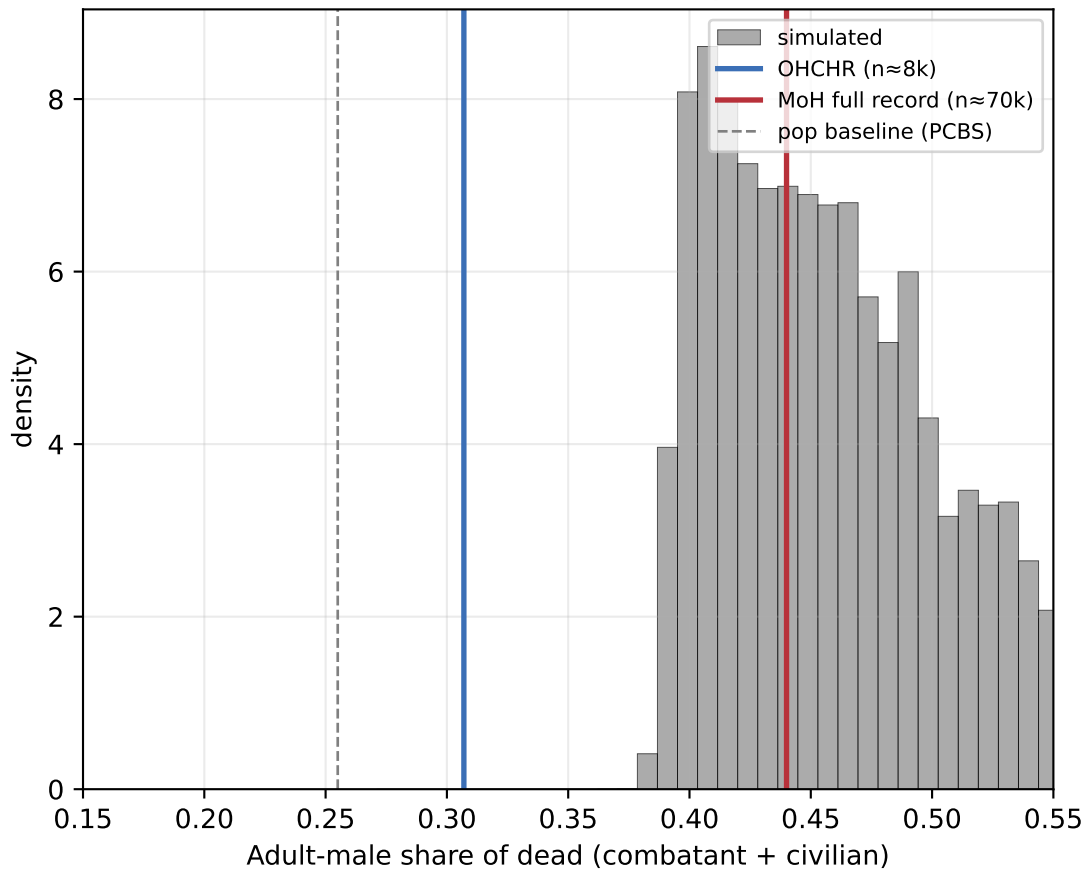
(6) Posterior decomposition of 70,100 reported deaths



(7) Implied militants by governorate (sum=47,720)



(8) Posterior fit to demographic data



Posterior (importance sampler, ESS=3,475)

Direct deaths in Gaza (true):
D_{total} median 81,890, 95% CI [66,412, 103,974]

Decomposition (median):

Militants killed	1,775
Adult-male civilians	30,249
Women + children	49,889

Militant share q median 2.2%
95% CI [1.5%, 3.0%]

Civilian:militant ratio (incl. kids):
median 45x, 95% CI [32x, 66x]

For comparison:
IDF/Israeli intel claim 17–25k militants killed
⇒ $q \approx 19\text{--}42\%$ (incompatible with posterior)
Naïve OHCHR-based Bayes $q \approx 6.0\%$ (CI 2.4–9.2%)
Simulator (this fig.) $q \approx 2.2\%$ (CI 1.5–3.0%)